MAT 3007 OPTIMIZATION
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MIDTERM EXAMINATION 26 SPRING

Mar. 21, 2026

Question	Points	Score
True/False	15	
The Simplex Method	21	
Duality	21	
Sensitivity Analysis	20	
Modeling	12	
Maximum Flow and Minimum Cut	12	
Total:	101	

- Please write down your **student ID** on the **answer paper**.
- Please justify your answers except Question 1.
- The exam time is 2 hours.
- Even if you are not able to answer all parts of a question, write down the part you know. You will get corresponding credits to that part.

Question 1 [15 points]: True/False

State whether each of the following statements is *True* or *False*. For each part, only your answer, which should be one of True or False, will be graded. Explanations are not required and will not be read.

(a) [3 points] A linear programming (LP) problem of standard form must have at least one basic feasible solution.

Solution: False. The problem can be infeasible.

(b) [3 points] If an LP problem has at least two distinct optimal solutions, then it has infinitely many distinct optimal solutions.

Solution: True.

(c) [3 points] If both an LP (primal) problem and its corresponding dual problem are feasible, then they both have optimal solutions and their optimal objective values are equal.

Solution: True.

(d) [3 points] A basic solution associated with the basis B of the standard form LP problem $\min \boldsymbol{c}^\top \boldsymbol{x}$ s.t. $\boldsymbol{A}\boldsymbol{x} = \boldsymbol{b},\ \boldsymbol{x} \geq \boldsymbol{0}$ is feasible if $\boldsymbol{A}^\top \boldsymbol{y} \leq \boldsymbol{c}$ for $\boldsymbol{y} = (\boldsymbol{A}_B^{-1})^\top \boldsymbol{c}_B$.

Solution: False. Primal feasibility of a basic solution is not implied by feasibility of the associated dual solution. Instead, if both solutions are feasible, then both are optimal (complementarity).

(e) [3 points] For an LP problem of standard form, if the solution value of a dual variable associated with a particular constraint is zero in an optimal dual solution, then changing the right-hand side of that constraint by an arbitrary amount cannot change the optimal objective value.

Solution: False. We need to additional assumptions: (i) the optimal dual solution is unique, and (ii) the right-hand side of that constraint is changed by a small enough amount.

Question 2 [21 points]: The Simplex Method

Consider the following LP problem:

$$\begin{array}{lll} \underset{x_1,x_2,x_3}{\text{maximize}} & 5x_1 + 4x_2 + 3x_3 & \\ \text{subject to} & 2x_1 + x_2 + x_3 & \leq 8 \\ & x_1 + 2x_2 + 3x_3 & \geq 6 \\ & x_1 + x_2 + x_3 & = 5 \\ & x_1, x_2, x_3 & \geq 0. \end{array}$$

(a) [5 points] Convert the problem to standard form.

Solution: Introducing slack variables s_1 and s_2 for the first two constraints, the standard form of the original LP problem is as follows

$$\begin{array}{lll} \underset{x_1,x_2,x_3}{\text{maximize}} & 5x_1 + 4x_2 + 3x_3 & \text{(1pts)} \\ \text{subject to} & 2x_1 + x_2 + x_3 + s_1 = 8 & \text{(1pts)} \\ & x_1 + 2x_2 + 3x_3 - s_2 = 6 & \text{(1pts)} \\ & x_1 + x_2 + x_3 = 5 & \text{(1pts)} \\ & x_1, x_2, x_3, s_1, s_2 \geq 0. & \text{(1pts)} \end{array}$$

(b) [6 points] Write down the Phase I problem and the initial Phase I tableau for the two-phase simplex method.

Solution: Introducing auxiliary variables y_1, y_2, y_3, the Phase I problem is of the form

$$\begin{aligned} \underset{x_1,x_2,x_3}{\text{minimize}} \quad & y_1 + y_2 + y_3 \\ \text{subject to} \quad & 2x_1 + x_2 + x_3 + s_1 + y_1 = 8 \\ & x_1 + 2x_2 + 3x_3 - s_2 + y_2 = 6 \\ & x_1 + x_2 + x_3 + y_3 = 5 \\ & x_1, x_2, x_3, s_1, s_2, y_1, y_2, y_3 \geq 0. \end{aligned}$$ (3pts)

The initial Phase I tableau is

B	−4	−4	−5	−1	1	0	0	0	-19
6	2	1	1	1	0	1	0	0	8
7	1	2	3	0	−1	0	1	0	6
8	1	1	1	0	0	0	0	1	5

. (3pts)

(c) [10 points] Apply the smallest index rule for pivoting, and solve the Phase I problem to optimality using simplex tableaus. Please list all intermediate tableaus and the final optimal solution of the Phase I problem.

Solution: Pivot 1: x_1 enters, y_1 leaves. After pivot:

B	0	−2	−3	1	1	2	0	0	-3
1	1	1/2	1/2	1/2	0	1/2	0	0	4
7	0	3/2	5/2	−1/2	−1	−1/2	1	0	2
8	0	1/2	1/2	−1/2	0	−1/2	0	1	1

(3pts)

Pivot 2: x_2 enters, y_2 leaves. After pivot:

B	0	0	1/3	1/3	−1/3	4/3	4/3	0	-1/3
1	1	0	−1/3	2/3	1/3	2/3	−1/3	0	10/3
2	0	1	5/3	−1/3	−2/3	−1/3	2/3	0	4/3
8	0	0	−1/3	−1/3	1/3	−1/3	−1/3	1	1/3

(3pts)

Pivot 3: s_2 enters, y_3 leaves. After pivot:

B	0	0	0	0	0	1	1	1	0
1	1	0	0	1	0	1	0	−1	3
2	0	1	1	−1	0	−1	0	2	2
5	0	0	−1	−1	1	−1	−1	3	1

(3pts)

The final optimal solution is $x_1^* = 3$, $x_2^* = 2$, $s_2^* = 1$, $x_3^* = s_1^* = y_1^* = y_2^* = y_3^* = 0$ (1pts).

Note: It is okay to not clarify which variable enters or leaves for each pivot, as long as the bases are correct. 3 points will be deducted if some pivoting rule other than the smallest index rule is applied and the final solution is correct.

Question 3 [21 points]: Duality

Consider the following linear programming problem:

$$\begin{array}{lll} \underset{x_1,x_2,x_3}{\text{minimize}} & x_1 + 2x_2 - 3x_3 & \\ \text{subject to} & x_1 + x_2 + x_3 & = 4 \\ & x_1 - x_2 & \geq 1 \\ & x_1, x_2, x_3 & \geq 0. \end{array}$$

(a) [7 points] Derive the dual problem.

Solution: The dual problem is given by

$$\begin{array}{lll} \underset{y_1,y_2}{\text{maximize}} & 4y_1 + y_2 & \text{(1pts)} \\ \text{subject to} & y_1 + y_2 \leq 1 & \text{(1pts)} \\ & y_1 - y_2 \leq 2 & \text{(1pts)} \\ & y_1 \leq -3 & \text{(1pts)} \\ & y_1 \text{ free}, y_2 \geq 0 & \text{(3pts)} \end{array}$$

(b) [7 points] Given that $\boldsymbol{x}^* = (1, 0, 3)$ is a feasible solution to the primal problem. Please use the **Strong Duality Theorem** and **Complementarity Condition** to determine the optimal solution to the dual problem, $\boldsymbol{y}^* = (y_1^*, y_2^*)$, and verify the optimal objective value.

Solution: First, we evaluate the primal objective function at $\boldsymbol{x}^* = (1, 0, 3)$:

$$Z^* = 1(1) + 2(0) - 3(3) = 1 - 9 = -8.$$

By the **Strong Duality Theorem**, if $\boldsymbol{x}^*$ is optimal, the optimal objective value of the dual problem must also be -8 (2pts).
To find the values of $\boldsymbol{y}^*$, we apply Complementary Slackness conditions:

i. Since $x_1^* = 1 > 0$, the corresponding first dual constraint must be binding (active):

$$y_1 + y_2 = 1 \quad \text{(1pts)}$$

ii. Since $x_3^* = 3 > 0$, the corresponding third dual constraint must be binding (active):

$$y_1 = -3 \quad \text{(1pts)}$$

Substituting $y_1 = -3$ into the first equation:

$$-3 + y_2 = 1 \implies y_2 = 4 \quad \text{(1pts)}$$

Thus, $\boldsymbol{y}^* = (-3, 4)$.
Finally, we verify feasibility and objective value (2pts):

- $y_2 = 4 \geq 0$ (Feasible).
- Check 2nd constraint: $y_1 - y_2 = -3 - 4 = -7 \leq 2$ (Feasible).
- Dual Objective: $4(-3) + (4) = -12 + 4 = -8$.

Since the dual objective equals the primal objective, Strong Duality is satisfied.

(c) [7 points] If we change the coefficient of x_2 from 2 to -4 in the primal objective function (i.e., the objective becomes "$x_1 - 4x_2 - 3x_3$"), show that $\boldsymbol{x}^* = (1, 0, 3)$ remains an optimal solution to the new primal problem.

Hint: You can first investigate the changes in the optimal solution to its dual.

Solution: For $\boldsymbol{x}^*$ to remain optimal, there must exist a dual feasible solution $\boldsymbol{y}$ that satisfies the complementary slackness conditions with $\boldsymbol{x}^*$.

Since the basis variables in $\boldsymbol{x}^*$ (x_1 and x_3) have not changed their cost coefficients, the system of equations derived from Complementary Slackness in part (b) remains the same:

$$y_1 + y_2 = 1 \quad \text{(Constraint for } x_1, c_1 = 1\text{)}$$
$$y_1 = -3 \quad \text{(Constraint for } x_3, c_3 = -3\text{)}$$

This yields the same dual candidate $\boldsymbol{y} = (-3, 4)$ (2pts).

We must now check if this $\boldsymbol{y}$ is feasible for the *new* dual problem. The only change in the dual problem is the constraint corresponding to x_2 (where the cost changed to $c_2' = -4$):

$$y_1 - y_2 \leq -4 \quad \text{(2pts)}$$

Checking our candidate $\boldsymbol{y} = (-3, 4)$:

$$\text{LHS} = (-3) - (4) = -7$$

$$\text{RHS} = -4$$

Since $-7 \leq -4$, the constraint is satisfied (2pts).

Since $\boldsymbol{x}^*$ is primal feasible, $\boldsymbol{y}$ is dual feasible, and they satisfy complementary slackness, $\boldsymbol{x}^*$ remains optimal (1pts).

Question 4 [20 points]: Sensitivity Analysis

Consider the following linear program:

$$\begin{array}{ll} \underset{\boldsymbol{x}}{\text{maximize}} & 3x_1 + 5x_2 \\ \text{subject to} & x_1 + 2x_2 \le 14, \\ & 3x_1 + x_2 \le 18, \\ & x_1 \le 4, \\ & x_1, x_2 \ge 0. \end{array} \tag{1}$$

The following gives the final simplex tableau when solving the standard form of problem (1).

B	0	0	5/2	0	1/2	37
2	0	1	1/2	0	−1/2	5
4	0	0	−1/2	1	−5/2	1
1	1	0	0	0	1	4

From the optimal simplex tableau, answer the following questions.

(a) [3 points] What is the optimal solution and the optimal value of problem (1)?

Solution: From the final simplex tableau, the basic variables are x_2, x_4, and x_1 with values

$$x_2 = 5, \qquad x_4 = 1, \qquad x_1 = 4.$$

Hence the optimal solution to the original problem is

$$\boldsymbol{x}^* = (x_1^*, x_2^*) = (4, 5), \quad \text{(2pts)}$$

and the optimal value is

$$z^* = 3 \times 4 + 5 \times 5 = 37. \quad \text{(1pts)}$$

(b) [6 points] In what range can we change the right-hand-side coefficient of the first constraint, namely $b_1 = 14$ in $x_1 + 2x_2 \le 14$, so that the current optimal basis of the standard linear program remains optimal?

Solution: Let the change in b_1 be λ. Then the new right-hand side is $\tilde{\boldsymbol{b}} = \boldsymbol{b} + \lambda \boldsymbol{e}_1$, where $\boldsymbol{e}_1 = (1, 0, 0)^\top$ is the first standard basis vector in $\mathbb{R}^3$.
From the tableau the current basis is $B = \{2, 4, 1\}$ and

$$\boldsymbol{A} = \begin{bmatrix} 1 & 2 & 1 & 0 & 0 \\ 3 & 1 & 0 & 1 & 0 \\ 1 & 0 & 0 & 0 & 1 \end{bmatrix}, \qquad \boldsymbol{A}_B = \begin{bmatrix} 2 & 0 & 1 \\ 1 & 1 & 3 \\ 0 & 0 & 1 \end{bmatrix}.$$

Thus

$$\boldsymbol{A}_B^{-1} = \begin{bmatrix} \frac{1}{2} & 0 & -\frac{1}{2} \\ -\frac{1}{2} & 1 & -\frac{5}{2} \\ 0 & 0 & 1 \end{bmatrix}.$$

The current basic solution is

$$\boldsymbol{x}_B^* = \boldsymbol{A}_B^{-1}\boldsymbol{b} = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix}.$$

Since only the right-hand side changes, the reduced costs remain unchanged. Therefore we only need feasibility of the basic variables:

$$\boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1}\boldsymbol{e}_1 \ge 0. \quad \text{(2pts)}$$

Now

$$\boldsymbol{A}_B^{-1}\boldsymbol{e}_1 = \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix}.$$

Hence

$$\boldsymbol{x}_B^* + \lambda \boldsymbol{A}_B^{-1}\boldsymbol{e}_1 = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix} + \lambda \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix} \geq 0. \quad \text{(1pts)}$$

This gives

$$5 + \frac{\lambda}{2} \geq 0, \qquad 1 - \frac{\lambda}{2} \geq 0,$$

so

$$-10 \leq \lambda \leq 2. \quad \text{(2pts)}$$

Since $b_1 = 14 + \lambda$, the allowable range is

$$4 \leq b_1 \leq 16. \quad \text{(1pts)}$$

(c) [5 points] If we change $b_1 = 14$ to $b_1 = 16$, what are the new optimal solution and the new optimal value?

Solution: If b_1 changes from 14 to 16, then $\lambda = 2$ and $\Delta \boldsymbol{b} = (2, 0, 0)^\top$. The new basic solution is

$$\tilde{\boldsymbol{x}}_B = \boldsymbol{A}_B^{-1}(\boldsymbol{b} + \Delta \boldsymbol{b}) = \boldsymbol{x}_B^* + \boldsymbol{A}_B^{-1}\Delta \boldsymbol{b}. \quad \text{(1pts)}$$

Thus

$$\tilde{\boldsymbol{x}}_B = \begin{bmatrix} 5 \\ 1 \\ 4 \end{bmatrix} + 2 \begin{bmatrix} \frac{1}{2} \\ -\frac{1}{2} \\ 0 \end{bmatrix} = \begin{bmatrix} 6 \\ 0 \\ 4 \end{bmatrix}. \quad \text{(2pts)}$$

Hence the new optimal solution is

$$\tilde{\boldsymbol{x}} = (4, 6), \quad \text{(1pts)}$$

and the new optimal value is

$$\tilde{z} = 3 \times 4 + 5 \times 6 = 42. \quad \text{(1pts)}$$

(d) [6 points] In what range can we change the objective coefficient 3 of variable x_1 so that the current optimal basis of the standard form linear program remains optimal?

Solution: Let the change in 3 be λ, so the new coefficient in standard form is $\tilde{c}_1 = -(3 + \lambda)$. Since the current basis is $B = \{2, 4, 1\}$, the variable x_1 is the third basic variable in the order (x_2, x_4, x_1). Hence we use $\boldsymbol{e}_3 = (0, 0, 1)^\top$.

To keep the current basis optimal, the reduced costs of nonbasic variables must remain nonnegative. The nonbasic variables are x_3 and x_5, and from the tableau

$$\boldsymbol{r}_N^\top = \left(\frac{5}{2}, \frac{1}{2}\right).$$

Also

$$\boldsymbol{A}_N = \begin{bmatrix} 1 & 0 \\ 0 & 0 \\ 0 & 1 \end{bmatrix}.$$

The new reduced costs are

$$\boldsymbol{r}_N^\top + \lambda \boldsymbol{e}_3^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N. \quad \text{(2pts)}$$

Since $\boldsymbol{e}_3^\top \boldsymbol{A}_B^{-1} = (0, 0, 1)$, we obtain

$$\boldsymbol{e}_3^\top \boldsymbol{A}_B^{-1} \boldsymbol{A}_N = (0, 1). \quad \text{(2pts)}$$

Therefore,

$$\left(\frac{5}{2}, \frac{1}{2}\right) + \lambda(0, 1) \geq \mathbf{0}^\top,$$

which gives

$$\frac{1}{2} + \lambda \geq 0.$$

Hence

$$\lambda \geq -\frac{1}{2}. \quad \text{(1pts)}$$

Since the new coefficient of x_1 in the original LP is $3 + \lambda$, the allowable range is

$$[\frac{5}{2}, +\infty). \quad \text{(1pts)}$$

Question 5 [12 points]: Modeling

You manage a fund that can invest 5 assets with expected returns $9\%, 7.5\%, 6\%, 10.5\%$ and 8%, respectively. Several portfolio rules must be respected:

- The total investment must sum to 100% (i.e., 1).
- The investment on each asset cannot exceed their individual allocation upper bounds (40% for asset 1, 35% for asset 2, 50% for asset 3, 25% for asset 4, 30% for asset 5).
- Asset 3 must be at least 20%.
- Average ESG score of the portfolio must be at least 75. (The ESG scores for these 5 assets are 60, 85, 90, 55, 70, respectively.)
- The current holdings of these 5 assets are $30\%, 10\%, 20\%, 25\%, 15\%$, respectively. Define $x^{(0)} = (0.3, 0.1, 0.2, 0.25, 0.15)$. The total (absolute) change in holding weights cannot exceed 40%.

Formulate a linear program that maximizes the expected return of the portfolio while respecting all portfolio rules.

Solution:

- Decision Variables: For $i = 1, \ldots, 5$, let x_i represent the fraction of investment in asset i and y_i represent the absolute change from the current holding of asset i. (2pts).
- The objective is to maximize the expected return, i.e., $0.09x_1 + 0.075x_2 + 0.06x_3 + 0.105x_4 + 0.08x_5$ (1pts).
- Constraints:
  - Sum-to-one constraint: $x_1 + x_2 + x_3 + x_4 + x_5 = 1$ (1pts);
  - Allocation upper bounds: $x_1 \leq 0.4$, $x_2 \leq 0.35$, $x_3 \leq 0.5$, $x_4 \leq 0.25$, $x_5 \leq 0.3$ (1pts);
  - Minimum holding for asset 3: $x_3 \geq 0.2$ (1pts);
  - ESG score constraint: $60x_1 + 85x_2 + 90x_3 + 55x_4 + 70x_5 \geq 75$ (2pts);
  - Defining y_i variables: $y_i \geq x_i - x_i^{(0)}$, $y_i \geq x_i^{(0)} - x_i$ for $i = 1, \ldots, 5$ (2pts);
  - Upper bounding absolute change: $y_1 + y_2 + y_3 + y_4 + y_5 \leq 0.4$ (1pts);
  - Nonnegativity constraints: $x_i \geq 0$, $i = 1, \ldots, 5$ (1pts).

Note: Other equivalent reformulations are also fine, e.g., using percentages rather than fractions to define variables (constraint coefficients should change accordingly). It is fine to define extra parameters, e.g., ESG scores or allocation upper bounds. It is fine to introduce extra nonnegativity constraints on y_i variables. If the upper bounding absolute change constraint is formulated as $\sum_{i=1}^{5} |x_i - x_i^{(0)}| \leq 0.4$, 2 points should be deducted.

Question 6 [12 points]: Maximum Flow and Minimum Cut

Consider a directed graph $G = (V, E)$, where $V = \{1, \ldots, n\}$ is the set of vertices, and E is a set of ordered tuples of the form (i, j) with $i, j \in V$ representing directed edges in the graph, with u_{ij} denoting the capacity of edge $(i, j) \in E$. The maximum flow problem seeks to find a flow from a source node $s \in V$ to the sink node $t \in V$ through the network that obtains the maximum possible flow rate while respecting the edge capacities. The LP formulation for maximum flow is of the form

$$
\begin{aligned}
F^* = \underset{F, \boldsymbol{f}}{\text{maximize}}\ & F \\
\text{subject to}\ & \sum_{j:(s,j)\in E} f_{sj} - \sum_{i:(i,s)\in E} f_{is} - F = 0 && \text{(define flow value at source } s\text{)} \\
& \sum_{j:(t,j)\in E} f_{tj} - \sum_{i:(i,t)\in E} f_{it} + F = 0 && \text{(define flow value at sink } t\text{)} \\
& \sum_{j:(v,j)\in E} f_{vj} - \sum_{i:(i,v)\in E} f_{iv} = 0, \qquad \forall v \in V \setminus \{s, t\} && \text{(flow conservation)} \\
& f_{ij} \le u_{ij}, \qquad \forall (i, j) \in E && \text{(edge capacity)} \\
& f_{ij} \ge 0, \qquad \forall (i, j) \in E.
\end{aligned}
$$

Here, f_{ij} represents the flow value on edge $(i, j) \in E$ and F represents the flow rate of the network. We assume there exists at least one directed path from s to t. Therefore, flow conservation constraints in the maximum flow LP imply that $\sum_{i:(i,t)\in E} f_{it} - \sum_{j:(t,j)\in E} f_{tj} = \sum_{j:(s,j)\in E} f_{sj} - \sum_{i:(i,s)\in E} f_{is}$.

(a) [7 points] Derive the dual LP of maximum flow. (You may assume the maximum flow problem is the dual maximization LP, and derive the corresponding primal minimization LP. **Hint:** You may start from a simple example and then generalize to general graphs.)

Solution: For $v \in V \setminus \{s, t\}$, define dual variable y_v for flow conservation at node v; let y_s and y_t denote the dual variable associated with the first constraint and the second constraint, respectively (1pts). For $(i, j) \in E$, define dual variable $z_{ij} \ge 0$ for edge capacity of edge (i, j) (1pts). The dual LP then takes the following form

$$
\begin{aligned}
\underset{\boldsymbol{y}, \boldsymbol{z}}{\text{minimize}}\ & \sum_{(i,j)\in E} u_{ij} z_{ij} \text{ (2pts)} \\
\text{subject to}\ & y_i - y_j + z_{ij} \ge 0, && \forall (i, j) \in E, \text{ (2pts)} \\
& -y_s + y_t = 1 \text{ (1pts)} \\
& z_{ij} \ge 0, && \forall (i, j) \in E.
\end{aligned}
$$

Note: Students get 2 points if they define all dual variables and their signs correctly, either explicitly like what we did, or directly in the formulation. Students get another 2 points if they get the formulation correct on an example.

(b) [5 points] An s-t cut $C = (S, T)$ of the graph G is a partition of the node set V into two disjoint subsets S and T such that $s \in S$, $t \in T$, $S \cap T = \emptyset$ and $V = S \cup T$. For example, in the case when $V = \{1, 2, 3, 4, 5\}$, $s = 1$, $t = 5$, $C = (S, T)$ is an s-t cut if defined by the following S and T:

- $S = \{1, 2, 3, 4\}$, $T = \{5\}$; or
- $S = \{1, 3, 4\}$, $T = \{2, 5\}$; or
- $S = \{1, 4\}$, $T = \{2, 3, 5\}$; or
- $\ldots$

The capacity $u(C)$ of the s-t cut $C = (S, T)$ is defined by the total capacity of edges going from the s side S to the t side T, i.e., $u(C) := \sum_{(i,j)\in E: i\in S, j\in T} u_{ij}$. Show that, for any s-t cut C, its capacity is an upper bound of the maximum flow, i.e., $u(C) \ge F^*$, using weak duality.

Solution: Given any s-t cut C, define $(\hat{y}, \hat{z})$ as follows:

- $\hat{y}_v = 0$ for $v \in S$, $\hat{y}_v = 1$ for $v \in T$; (1pts)
- $\hat{z}_{ij} = \hat{y}_j - \hat{y}_i$ for $(i, j) \in E$. (1pts)

Then $(\hat{y}, \hat{z})$ is feasible for the LP dual of maximum flow (1pts) with $u(C) = \sum_{(i,j)\in E} u_{ij}\hat{z}_{ij}$ (1pts). Therefore, by weak duality of LP, $u(C) \geq F^*$ (1pts).